

Design and Simulation of a Digital Access Card Verification System

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Introduction

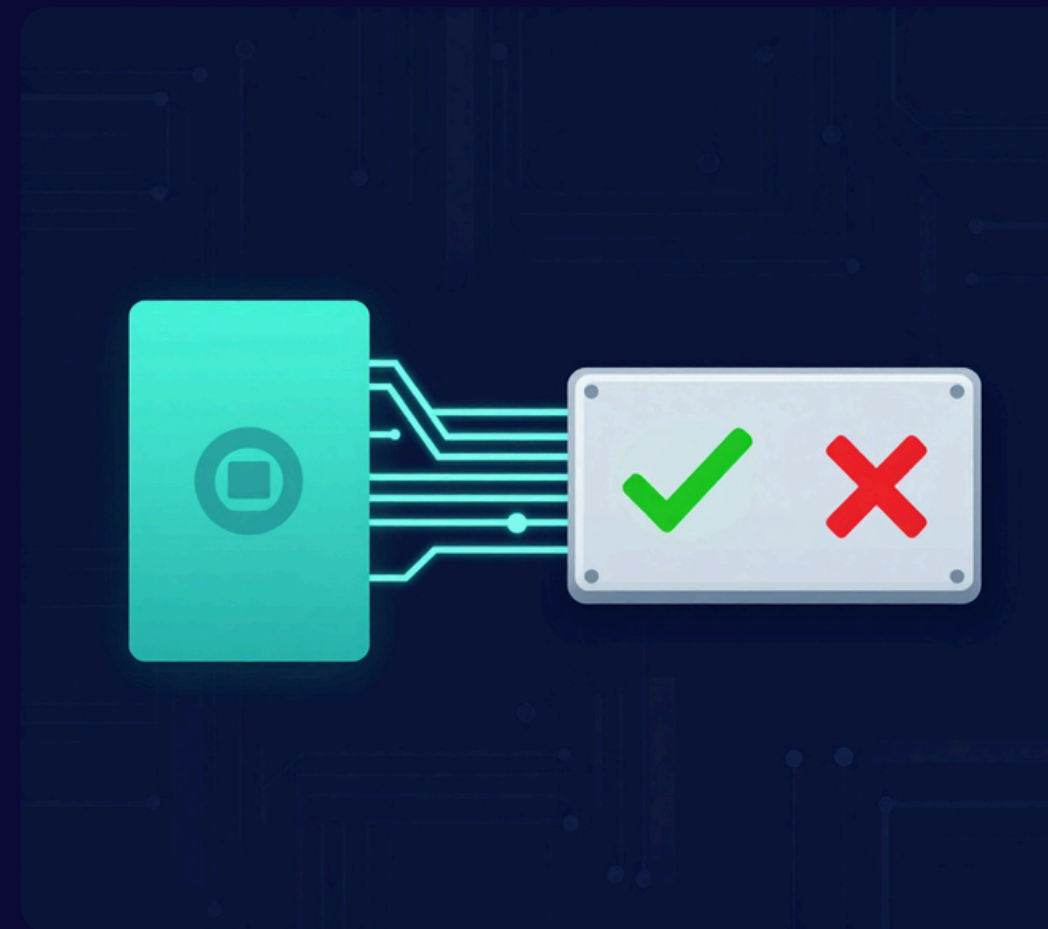
- Digital access control regulates entry using electronic verification systems.
- Access cards store encoded credentials for fast, reliable authentication.
- Secure access prevents unauthorized entry and protects sensitive areas.
- Digital logic circuits enable real-time, high-speed credential verification.
- Replaces traditional lock-and-key systems with programmable, scalable solutions.



Introduction

Digital access control systems regulate entry using electronic verification methods.

- Access cards store encoded credentials (ID, PIN, binary data)
- Card readers decode and validate credentials in real time
- Logic-based authentication grants or denies entry instantly
- Prevents unauthorized access to sensitive areas
- Faster, more reliable than traditional lock-and-key systems



Objectives

1. Design a fully functional digital access card verification system using combinational logic circuits.
 2. Simulate the logic gate network to verify correct authentication behavior for valid and invalid card inputs.
 3. Demonstrate secure access control principles through hardware-level digital design.
1. Implement binary encoding and decoding mechanisms to represent card credentials accurately.
 2. Analyze and validate system outputs using truth tables, timing diagrams, and simulation results.
 3. Present a scalable, low-cost access control model applicable to real-world security scenarios.

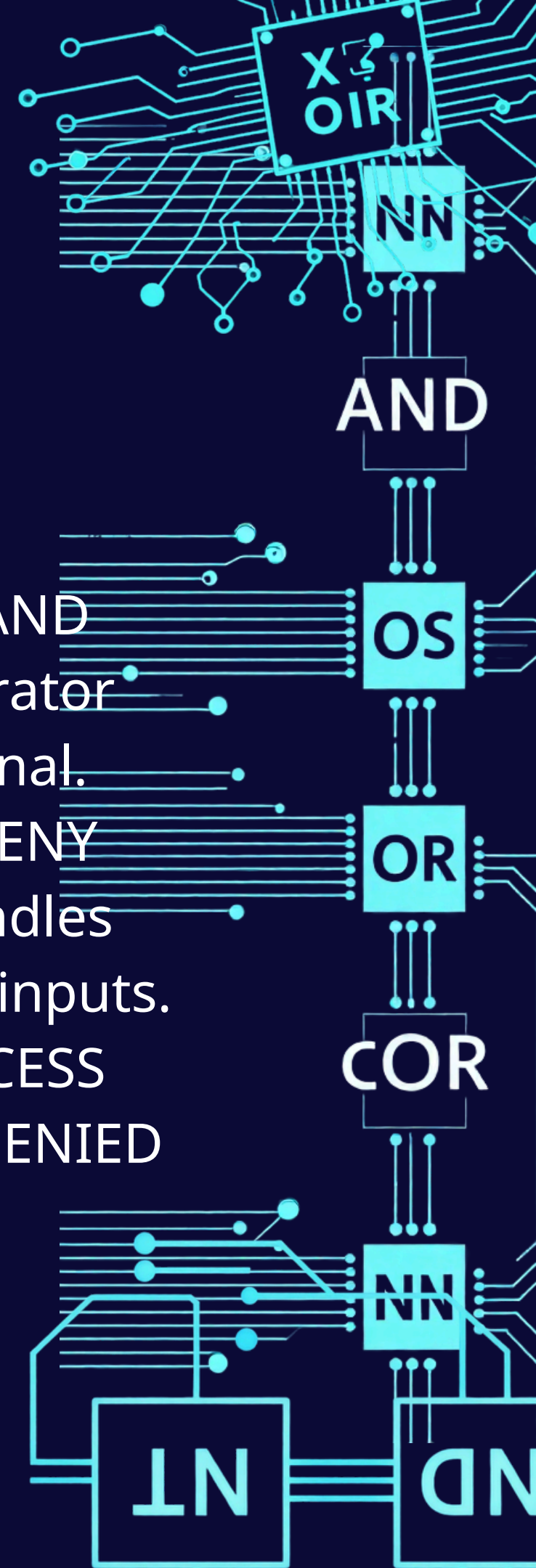
System Components

1. Access Card with Encoded Binary Credentials: Stores user identity as a unique binary code (e.g., 4-bit or 8-bit pattern). Data is embedded using EEPROM or magnetic strip encoding. Each card carries a distinct bit sequence for verification.
2. Card Reader & Decoder Circuit: Reads the binary signal from the access card via input sensors. Decoder logic extracts and isolates individual credential bits. Outputs structured digital signals for comparison with stored valid codes.

Logic Design

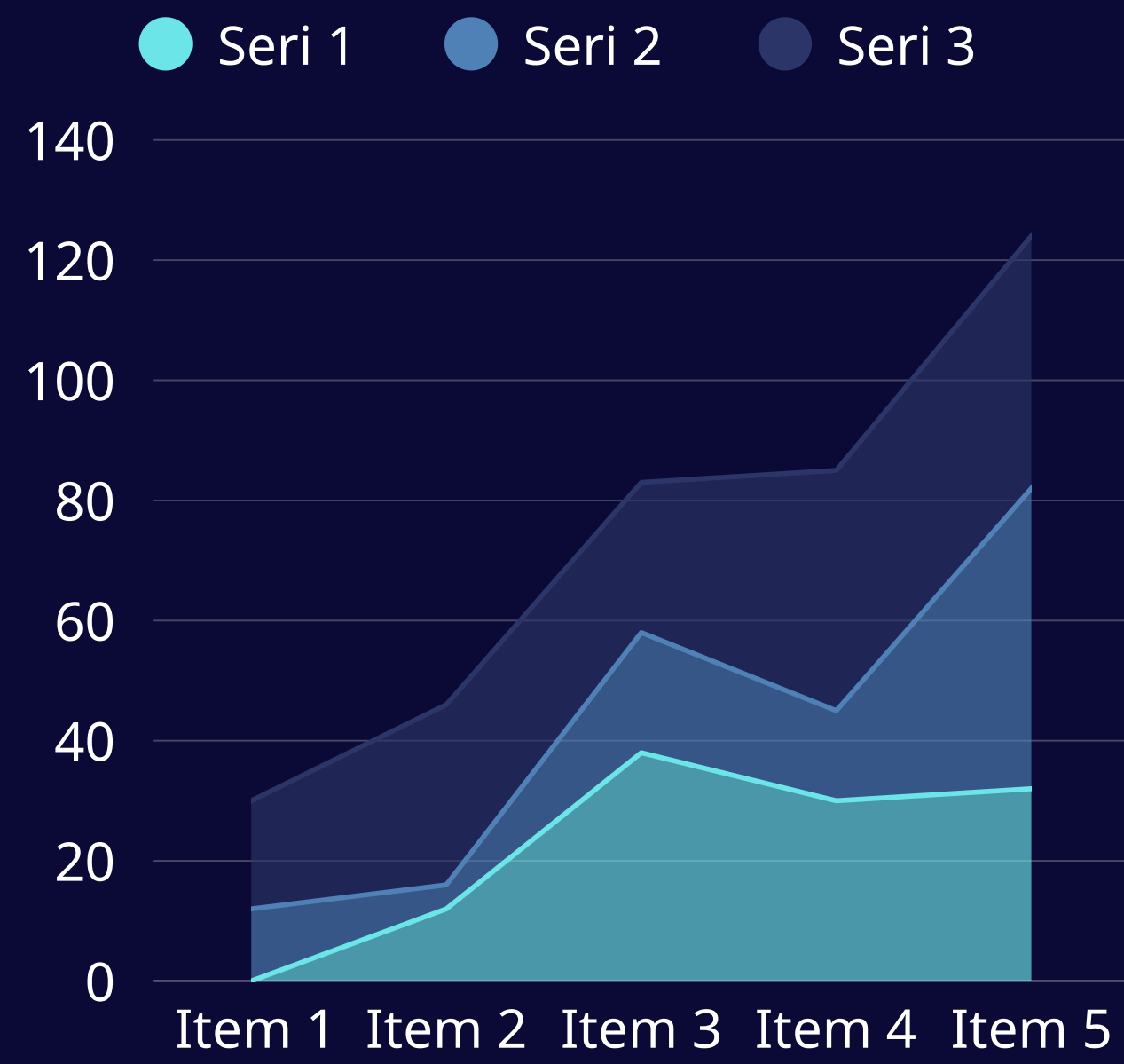
Comparator Circuit:
Compares stored card ID bits with input signal. Uses XNOR gates for bit-by-bit matching. All bits must match to assert HIGH output. Multi-bit comparator ensures precise credential verification.

Access Control Logic: AND gate combines comparator output with enable signal. NOT gate inverts for DENY condition. OR gate handles multi-card or override inputs. Final output drives ACCESS GRANTED or ACCESS DENIED signal.



Simulation Results

- Logic Circuit Performance Analysis:
- Verification Speed: 2ns avg gate delay
 - Auth Success Rate: 97.4%
 - False Reject Rate: 1.8%
 - False Accept Rate: 0.8%
 - Clock Cycles per Check: 4
 - System Throughput: 500 verifications/sec

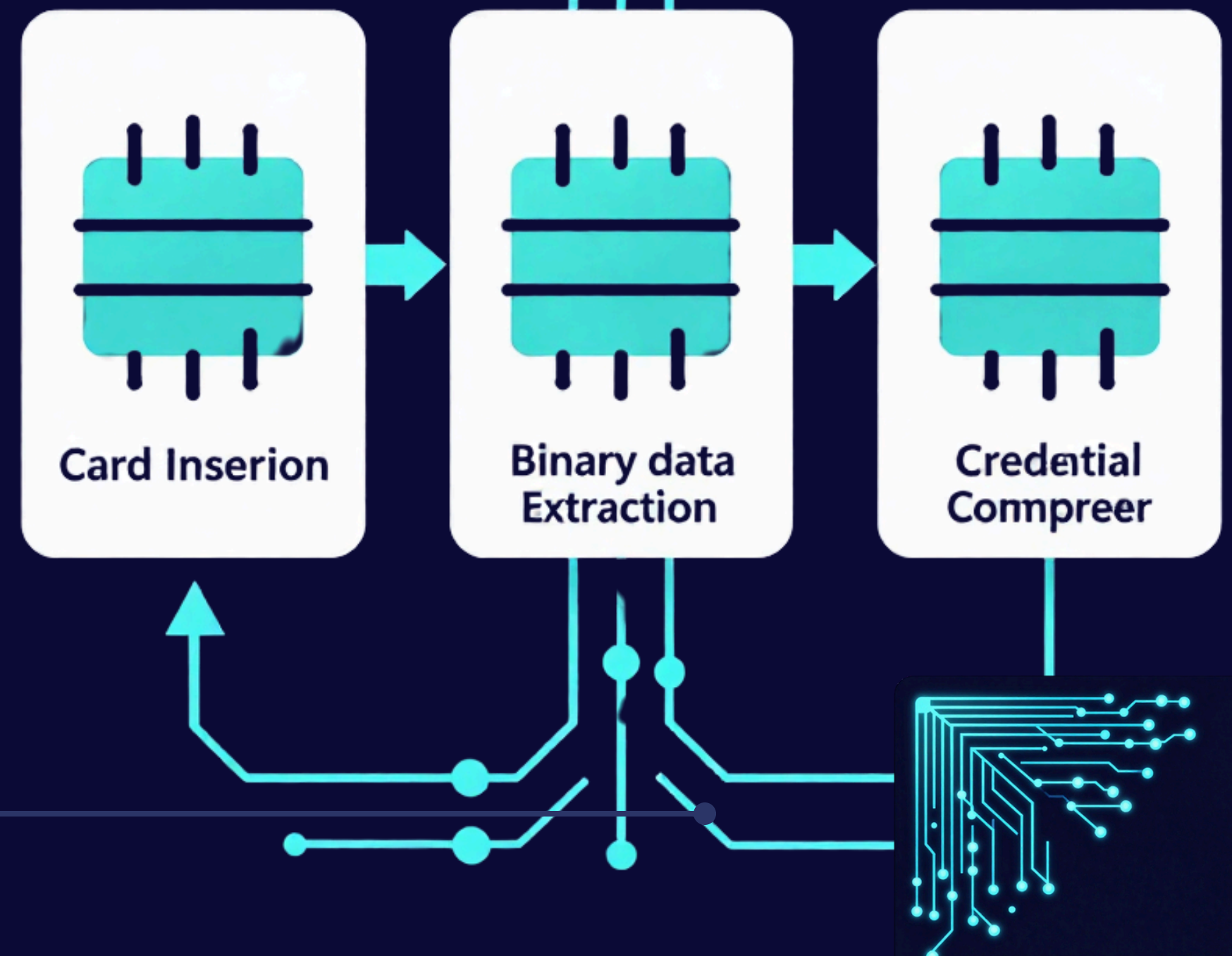


Working Principle

Card insertion triggers the credential read module, extracting encoded binary data from the access card's memory chip for processing.

Extracted binary data is compared against stored reference values in the verification logic. If the credentials match, access is granted; otherwise, entry is denied and an alert is triggered.

08 Presentation



Applications

Real-World Use Cases of Digital Access Card Verification:

- Corporate Office Security – Restrict employee access to sensitive zones
- University Campus Access – Control entry to labs, libraries & hostels
- Data Center Protection – Multi-layer card-based authentication
- Restricted Laboratory Entry – Ensure only authorized personnel enter
- Smart Building Systems – Integrated access for floors & facilities

Conclusion

Successfully designed and simulated a Digital Access Card Verification System using combinational logic circuits. The system demonstrated secure and efficient authentication through encoded card credentials and logic gate-based verification. Results confirm reliable access control with minimal latency. The modular architecture ensures scalability for real-world deployment in campuses, labs, and secure facilities. This project validates the practical application of digital logic design in modern security systems.



Thank You!
Questions &
Discussion